

Three-Bit Flows and Cycle Covers

Part I

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Abstract

We study the relationship between nowhere-zero three-bit flows, labeled triangles, and cycle double covers of bridgeless multigraphs. After reducing to the cubic case and using the flow theorems of Seymour and Tutte, we realize the three flow values at each vertex as the side differences of a triangle whose sides carry two-element subsets of $\mathbb{F}_2^3$. We express agreement of these local labels across graph edges as a binary system and prove its solvability by an inconsistency certificate, a local tester-parity identity, and a global double count. The resulting compatible labels trace cycle components in which every edge occurs exactly twice, proving that every finite bridgeless multigraph has a cycle double cover, *proving the cycle double cover conjecture*.

1 Introduction

The cycle double cover conjecture is a central and long-standing problem in graph theory: is the absence of bridges the only obstruction to covering every edge exactly twice by cycles? The conjecture was formulated independently by Szekeres and Seymour [7, 5]; related circuit-cover questions were studied by Itai and Rodeh [3]. It remained open in full generality despite extensive work on special classes, flows, and surface embeddings [4, 11, 2]. In this paper we prove this conjecture.

All graphs in this paper are finite undirected multigraphs. Parallel edges and loops are allowed unless explicitly excluded. A loop contributes two to the degree of its incident vertex. For an edge e , the notation $G - e$ means that e is deleted while its endpoints are retained. An edge e is a *bridge* if $G - e$ has more connected components than G ; a graph is *bridgeless* if it has no bridge. A loop is never a bridge. Isolated vertices have no effect on cycle double covers. For a vertex v , we write $\delta(v)$ for the set of edges incident with v ; because loops are excluded whenever this notation is used below, each such edge occurs once in $\delta(v)$.

Definition 1.1. A *cycle* is a connected 2-regular subgraph. Thus a loop is a cycle, and two parallel edges joining the same two vertices form a cycle of length two. In particular, a cycle uses each of its edges once, although a vertex of the ambient multigraph may be incident with additional edges not belonging to the cycle.

Definition 1.2. A *cycle double cover* of a graph G is a finite multiset $\mathcal{C}$ of cycles such that every edge of G belongs to exactly two members of $\mathcal{C}$, counted with multiplicity. Equal cycles may occur more than once in $\mathcal{C}$; their occurrences are regarded as distinct members.

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The empty multiset is a cycle double cover of an edgeless graph. Adding or deleting isolated vertices changes neither the cycles nor the edge-covering condition, which justifies ignoring isolated vertices throughout.

A loop is never a bridge. If a non-loop edge lies on a cycle, then after the edge is deleted the remaining part of the cycle still joins its two endpoints; such an edge is not a bridge either. In a cycle double cover every edge lies on a cycle, so no edge can be a bridge. Thus bridgelessness is necessary, and the cycle double cover conjecture asserts that it is also sufficient.

Conjecture 1.3 (Cycle Double Cover Conjecture). *Every finite bridgeless multigraph has a cycle double cover.*

1.1 Related work

Szekeres formulated the problem through polyhedral decompositions of cubic graphs [7], while Seymour’s circuit formulation [5] and the work of Itai and Rodeh [3] placed it in the now-standard language of circuit and cycle covers. Jaeger’s survey [4] describes the early theory and the reduction to cubic graphs; Zhang’s monograph [11] gives a comprehensive account of circuit double covers, reductions, and related conjectures.

Several important classes were previously known to satisfy the conjecture. For a bridgeless plane graph, the facial boundary walks, after decomposition into cycles, give a cycle double cover [4]. A properly 3-edge-colored cubic graph also has an immediate cycle double cover: the union of each pair of color classes is a spanning 2-regular subgraph, and each color class occurs in two of the three pairs. Alspach, Goddyn, and Zhang established the conjecture for the broad class of bridgeless graphs containing no subdivision of the Petersen graph [1]. More recently, the surface-embedding formulation has motivated quantitative work on embeddings with few singular edges [2].

Nowhere-zero flows form another major line of attack. Tutte’s group-flow theorem relates integer flows to flows over finite abelian groups of the same order [8], and Seymour proved that every bridgeless graph has a nowhere-zero integer 6-flow [6]. The latter is also an integer 8-flow; applying Tutte’s theorem with the group $\mathbb{F}_2^3$ therefore gives a nowhere-zero three-bit flow. These theorems supply the established external input to our argument. The new step is a global compatibility theorem that converts the three-bit flow into two-element edge labels satisfying the local cycle condition. This conversion is the mechanism that completes the cycle double cover.

Connections between low-order flows and cycle double covers have been studied in several forms. Zhang related nowhere-zero 4-flows to cycle double covers with bounded numbers of cycles and proved further extension results [10]. Xie and Zhang introduced flow-pair covers and used suitable pairs of integer flows to obtain cycle double covers under additional parity and support hypotheses [9]. Those results emphasize that a flow by itself does not automatically display the required cycles. The flow-to-cover theorem proved here uses a different mechanism: for a cubic multigraph, an arbitrary nowhere-zero $\mathbb{F}_2^3$ -flow is converted into globally compatible two-element edge labels, from which the cycle components are read off directly.

As a basic example, suppose that a loopless cubic multigraph has a proper 3-edge-coloring. Every vertex is incident with one edge of each color. The union of any two color classes is therefore a spanning 2-regular subgraph, hence a disjoint union of cycles. There are three choices of two color classes. An edge of one fixed color belongs to exactly the two choices containing that color, so the cycle components of these three 2-factors form a cycle double cover. Thus the non-3-edge-colorable cubic multigraphs constituted the essential unresolved case after the cubic reduction.

1.2 Proof strategy and roadmap

Our proof converts a nowhere-zero three-bit flow into a cycle double cover by passing through a system of locally defined, globally compatible two-element labels. The argument has six stages.

1. *Structural reduction.* Section 2 establishes the elementary cycle and trail facts needed for the reduction. Proposition 3.1 in Section 3 replaces an arbitrary bridgeless multigraph by a loopless bridgeless cubic multigraph in such a way that a cycle double cover of the cubic graph projects to a cycle double cover of the original graph. It therefore suffices to prove the conjecture in the cubic case.
2. *Three-bit flow input.* Section 4 combines Seymour's 6-flow theorem (Theorem 4.3) with Tutte's group-flow theorem (Theorem 4.4). Their consequence, Corollary 4.5, supplies a nowhere-zero $\mathbb{F}_2^3$ -flow f . At a cubic vertex its three incident values are distinct nonzero vectors whose XOR is zero, as recorded in Lemma 4.6.
3. *Local triangle labels.* In Section 5, the three flow values at each vertex are realized as the XOR differences of the sides of a triangle. A translation $t_v \in \mathbb{F}_2^3$ moves all three triangle labels simultaneously. Lemma 5.1 proves the crucial local invariant: every label occurs on either zero or two sides of the triangle. Lemma 5.2 expresses every side in a form suitable for comparison across a graph edge.
4. *The global gluing system.* Section 6 asks for translations t_v that make the two endpoint triangles assign the same unordered label pair to each edge. Proposition 6.2 shows that this is equivalent to the binary system

$$t_u \oplus t_v \oplus \epsilon_e f(e) = d_e \quad (e = uv).$$

Thus the geometric compatibility question becomes a finite system of linear equations over $\mathbb{F}_2$.

5. *Elimination of compatibility obstructions.* Section 7 uses Lemma 7.1 to represent any inconsistency by an XOR certificate $0 = 1$. Lemma 7.3 encodes such a certificate by one tester vector on each edge. The local calculation in Section 8, especially Lemma 8.1, identifies the discrepancy at a vertex with the parity of its incident nonzero testers. In Section 9, every nonzero edge tester is counted once at each endpoint. The resulting incidence count is even, so the global discrepancy is zero in $\mathbb{F}_2$. This contradicts the required value one in an inconsistency certificate and proves the triangle compatibility theorem, Theorem 9.1.
6. *Extraction of the cycle double cover.* Section 10 assigns to each label $s \in \mathbb{F}_2^3$ the subgraph consisting of the edges whose common two-element label set contains s . Lemma 10.1 uses triangle parity to show that every vertex has degree zero or two in this subgraph, so every edge-containing component is a cycle. The common two-element label set on each edge contains exactly two distinct labels and hence places that edge in exactly two indexed cycle components. Corollary 10.3 proves the cubic case, and Proposition 3.1 then proves Conjecture 1.3 in full.

Theorems 4.3 and 4.4 are the two external inputs. The principal new result is Theorem 9.1; all reductions and all subsequent steps converting the resulting three-bit flow into a cycle double cover are proved here. The resulting flow-to-cover statement is isolated as Theorem 10.2.

2 Preliminaries

A *trail* is a sequence $v_0e_1v_1e_2\cdots e_mv_m$ in which each edge e_i joins v_{i-1} to v_i and the edge objects $e_1, \dots, e_m$ are pairwise distinct. It is *closed* if $v_m = v_0$. Parallel edges are distinct edge objects. To say that a trail decomposes into edge-disjoint cycles means that the cycles have pairwise disjoint edge sets and that the union of those edge sets is exactly the set of edges in the trail.

Lemma 2.1 (Non-bridges lie on cycles). *Let G be a loopless multigraph and let $e = uv$ be an edge. Then e is not a bridge if and only if e lies on a cycle.*

Proof. Suppose first that e lies on a cycle D . Since G is loopless, u and v are distinct. Deleting e from D leaves a u - v path in $G - e$. Let x and y be vertices in the same component of G , and choose an x - y path Q in G . If Q avoids e , it remains in $G - e$. If Q uses e , replace that occurrence of e by the u - v path $D - e$. The result is an x - y walk in $G - e$, and every walk contains a path between its endpoints. Thus deleting e does not separate any two previously connected vertices and cannot increase the number of components. Hence e is not a bridge.

Conversely, suppose e is not a bridge. If u and v belonged to different components of $G - e$, then deleting e would split the component of G containing e into at least two components, so e would be a bridge. Therefore u and v lie in the same component of $G - e$. Choose a u - v path P in $G - e$. The union of P and e is connected, and every vertex in that union has degree two. It is therefore a cycle containing e . This argument also covers parallel edges: if P consists of one edge parallel to e , then $P \cup \{e\}$ is a cycle of length two. $\square$

Lemma 2.2 (A closed trail splits into cycles). *Every nonempty closed trail in a loopless multigraph decomposes into edge-disjoint cycles; equivalently, its edge set is the disjoint union of the edge sets of those cycles.*

Proof. We use strong induction on the number m of edges in the trail. Write the closed trail as

$$W = v_0e_1v_1e_2\cdots e_mv_m, \quad v_m = v_0,$$

where $m \geq 1$ and the edges $e_1, \dots, e_m$ are pairwise distinct, and assume the assertion holds for every nonempty closed trail with fewer than m edges. If no vertex is repeated except $v_0 = v_m$, then the subgraph traced by W is connected and 2-regular. Hence that subgraph is a cycle, and its edge set is the edge set of W .

Otherwise choose indices $0 \leq i < j \leq m$ with $v_i = v_j$ and with $j - i$ as small as possible among all repeated-vertex pairs. The segment

$$C = v_ie_{i+1}v_{i+1}\cdots e_jv_j \tag{2.1}$$

is a closed trail. It has no repeated internal vertex: any such repetition, or any internal occurrence of v_i , would give a repeated-vertex pair with smaller index difference. Moreover, $j - i \geq 2$, since $j - i = 1$ would make e_j a loop. Every internal vertex of the segment is incident in the traced subgraph with its entering and leaving edges, while $v_i = v_j$ is incident with the distinct first and last edges. Thus the traced subgraph is connected and 2-regular, so it is a cycle. We denote this cycle subgraph also by C .

Delete the segment (2.1) from the displayed order for W and concatenate the remaining portions at the common vertex $v_i = v_j$. The resulting sequence is

$$W' = v_0e_1\cdots e_iv_i(=v_j)e_{j+1}\cdots e_mv_m. \tag{2.2}$$

The initial portion is omitted when $i = 0$, and the final portion is omitted when $j = m$. If no edge remains, then C uses every edge of W and we are done. Otherwise (2.2) starts and ends at $v_0 = v_m$ and uses pairwise distinct edges, so it is a closed trail. It has $m - (j - i) < m$ edges. By the induction hypothesis, W' decomposes into edge-disjoint cycles. Those cycles are also edge-disjoint from C , because W' uses exactly the edges of W not used by C . Together with C , they give the required decomposition of W . $\square$

3 Reduction to cubic graphs

Proposition 3.1 (Cubic reduction). *Suppose every loopless bridgeless cubic multigraph has a cycle double cover. Then every finite bridgeless multigraph has a cycle double cover.*

Proof. The construction may be carried out simultaneously in all connected components. Equivalently, one may apply it componentwise and take the multiset union of the resulting covers. We proceed in four steps.

Step 1: remove loops. Let L be the set of loop edges of G , and let $G_0 = G - L$. A loop does not join two distinct vertices and cannot occur on a path between distinct vertices. Consequently, for every non-loop edge e and every pair of vertices x, y , the vertices x and y are connected in $G - e$ if and only if they are connected in $G_0 - e$. Thus $G - e$ and $G_0 - e$ have the same connected components, and e is a bridge of G if and only if it is a bridge of G_0 . Since G is bridgeless, G_0 is loopless and bridgeless.

If G_0 has no edges, begin with the empty cycle double cover of G_0 and skip Steps 2–4, proceeding directly to restoration of the loops at the end of the proof. Hence assume that G_0 has an edge. Isolated vertices of G_0 may be ignored. Every remaining vertex has degree at least two: if e were incident with a vertex of degree one, deleting e would isolate that vertex from the other endpoint of e , making e a bridge.

Step 2: expand every vertex into a cycle. For each nonisolated vertex v of G_0 , let $d = d_{G_0}(v) \geq 2$ and replace v by a new cycle, taking these replacement cycles to be pairwise vertex-disjoint:

$$R_v = v_1 v_2 \cdots v_d v_1.$$

When $d = 2$, this means two parallel edges joining v_1 and v_2 . Attach the d edge-ends incident with v bijectively to the d vertices of R_v . More formally, choose a bijection from the incidences (v, e) with $e \in \delta_{G_0}(v)$ to $V(R_v)$. If $e = uv$ is an edge of G_0 , join the chosen attachment vertex in R_u to the chosen attachment vertex in R_v . Denote this new edge by $\hat{e}$.

Call the edges $\hat{e}$ *old edges* and the edges on the cycles R_v *replacement edges*. Let H be the resulting graph. Each vertex of H is incident with two replacement edges and exactly one old edge, so H is cubic. It is loopless: replacement edges lie on loopless cycles R_v , including the length-two case, and an old edge $\hat{e}$ joins R_u to R_v for distinct original endpoints $u \neq v$ because G_0 is loopless.

Step 3: prove that H is bridgeless. Every replacement edge lies on the cycle R_v containing it. Now consider an old edge $\hat{e}$ corresponding to $e \in E(G_0)$. Since G_0 is loopless and bridgeless, Lemma 2.1 gives a cycle D of G_0 containing e .

We lift D to H . Include $\hat{a}$ for every edge $a \in E(D)$. At a vertex v of D , exactly two distinct edges of D are incident with v . Their edge-ends were attached to two distinct vertices of R_v , because the attachment map is a bijection. The cycle R_v is the union of two edge-disjoint paths between those attachment vertices. Choose either path and include all its replacement edges. We

call the chosen path an arc of R_v . In the length-two case, the two arcs are the two individual parallel replacement edges, so the same description applies.

The resulting subgraph is connected: it is obtained from the connected cycle D by replacing each vertex with a joining path. At an internal vertex of a chosen arc, exactly two chosen replacement edges are incident. At an endpoint of a chosen arc, exactly one chosen replacement edge and one chosen old edge are incident. No other vertices are used. Hence every vertex of the lifted subgraph has degree two. The lift is therefore a cycle of H , and it contains $\hat{e}$. Thus every edge of H lies on a cycle. By Lemma 2.1, H is bridgeless.

Step 4: contract a cover of H back to G . By the hypothesis of the proposition, H has a cycle double cover $\mathcal{D}$. This hypothesis applies whether or not H is connected. Define a projection from H to G_0 by contracting every replacement cycle R_v to the original vertex v , sending each old edge $\hat{e}$ to its original edge e , and suppressing all replacement edges.

Consider one occurrence of a cycle Z in the multiset $\mathcal{D}$. If Z uses no old edge, then its connectedness forces it to lie entirely in one replacement cycle; its projection is a single vertex, so discard it. Suppose instead that Z uses an old edge $\hat{e}$ with endpoints in distinct replacement cycles. If this were the only old edge of Z , then deleting $\hat{e}$ from Z would leave a path between its endpoints using only replacement edges. This is impossible because distinct replacement cycles are distinct components of the replacement-edge subgraph. Hence Z uses at least two old edges. Read them in their cyclic order. Each vertex of H is incident with only one old edge, so between two successive old edges, Z follows a nonempty path of replacement edges within a single cycle R_v . After contraction, the two old edges are therefore incident with the same original vertex v . The cyclic sequence of old edges consequently becomes a closed walk in G_0 . It may visit an original vertex more than once if Z passes through the corresponding replacement cycle in more than one segment.

No old edge occurs twice in this walk, because Z is a cycle and hence uses each edge of H at most once. The closed walk is therefore a closed trail. By Lemma 2.2, decompose it into edge-disjoint cycles of G_0 , and insert all those cycles into a new multiset $\mathcal{C}_0$. Carry out this operation separately for every occurrence of every member of $\mathcal{D}$; if equal cycles result from different occurrences, retain their multiplicity.

Each edge $e \in E(G_0)$ has exactly one old representative $\hat{e}$ in H . The edge $\hat{e}$ occurs in exactly two members of $\mathcal{D}$, counted with multiplicity. Each occurrence contributes one occurrence of e to exactly one cycle produced by the corresponding trail decomposition. No other operation creates or deletes an old-edge occurrence. Hence every edge of G_0 occurs in exactly two members of $\mathcal{C}_0$, so $\mathcal{C}_0$ is a cycle double cover of G_0 . The isolated vertices ignored in Step 2 have no incident edges and impose no additional covering condition.

Finally, insert two copies of each loop in L . Each loop is itself a cycle, and no cycle of G_0 contains a deleted loop. Thus every restored loop and every non-loop edge of the original graph is covered exactly twice. The result is a cycle double cover of G . $\square$

From now on, G denotes a loopless bridgeless cubic multigraph.

4 Three-bit colors and flows

Let $\mathbb{F}_2 = \{0, 1\}$ be the field with arithmetic modulo 2. The set

$$\{0, 1\}^3 = \{000, 001, 010, 011, 100, 101, 110, 111\}$$

is the three-dimensional vector space $\mathbb{F}_2^3$. We write its addition as $\oplus$ and call it *XOR*. The zero vector is 000, and addition is performed coordinate by coordinate modulo 2. For example, $100 \oplus 010 = 110$, $101 \oplus 101 = 000$. Thus every $s \in \{0, 1\}^3$ satisfies $s \oplus s = 000$, or equivalently $-s = s$.

Definition 4.1. Let A be an abelian group, written additively, and choose an orientation of each edge of a graph. Interpret an empty sum as the identity element $0 \in A$. An A -flow is a map $f: E(G) \rightarrow A$ such that, at every vertex v ,

$$\sum_{\substack{e \in E(G) \\ e \text{ directed out of } v}} f(e) = \sum_{\substack{e \in E(G) \\ e \text{ directed into } v}} f(e), \quad (4.1)$$

It is *nowhere-zero* if $f(e) \neq 0$ for every edge e . A *three-bit flow* is an A -flow with $A = \mathbb{F}_2^3$; in that case the sums in (4.1) are XOR sums. If loops are present, an oriented loop has its tail and head at the same vertex and therefore contributes once to each side of (4.1). In the main argument, however, G is loopless by the standing assumption following Proposition 3.1.

Definition 4.2. An *integer flow* is an orientation of the edges together with a map $\varphi: E(G) \rightarrow \mathbb{Z}$ satisfying the conservation equation (4.1) with ordinary integer addition. For an integer $k \geq 2$, it is an *integer k -flow* if $|\varphi(e)| < k$ for every edge e , and it is *nowhere-zero* if $\varphi(e) \neq 0$ for every edge e . Thus the values of a nowhere-zero integer k -flow belong to $\{\pm 1, \dots, \pm(k-1)\}$.

For a flow with values in any abelian group, reversing one oriented edge and replacing its value by its additive inverse preserves every conservation equation. Consequently, existence of a nowhere-zero flow is independent of the chosen orientation.

For $A = \mathbb{F}_2^3$, every value is its own additive inverse, so an edge value does not even have to be changed when its orientation is reversed. More directly, XORing the incoming side of (4.1) onto both sides gives

$$\left(\bigoplus_{e \text{ directed out of } v} f(e) \right) \oplus \left(\bigoplus_{e \text{ directed into } v} f(e) \right) = 000.$$

Because G is loopless, the edges incident with v are partitioned by these two indexing sets. Hence (4.1) is equivalent to

$$\bigoplus_{e \in \delta(v)} f(e) = 000. \quad (4.2)$$

At a cubic vertex, equation (4.2) is the XOR-zero rule for its three incident edge values.

We use two standard theorems as black boxes.

Theorem 4.3 (Seymour's 6-flow theorem [6]). *Every bridgeless graph has a nowhere-zero integer 6-flow.*

Theorem 4.4 (Tutte's group-flow theorem [8, 11]). *Let $k \geq 2$, and let A be any finite abelian group of order k . A finite graph has a nowhere-zero A -flow if and only if it has a nowhere-zero integer k -flow.*

A nowhere-zero integer 6-flow is also a nowhere-zero integer 8-flow: the same orientation and the same edge values satisfy $0 < |\varphi(e)| < 6 < 8$ for every edge e . Since $\mathbb{F}_2^3$ is an abelian group of order eight, the two theorems give the following input.

Corollary 4.5 (Three-bit flow input). *Every bridgeless graph has a nowhere-zero three-bit flow.*

Proof. Use Theorem 4.3 to obtain a nowhere-zero integer 6-flow, regard it as an integer 8-flow, and apply Theorem 4.4 with $k = 8$ and $A = \mathbb{F}_2^3$. $\square$

Lemma 4.6 (The local Fano-line rule). *Let f be a nowhere-zero three-bit flow on a loopless graph G , let v be a cubic vertex, and let x, y, z be the values of the three edges incident with v , listed in any order. Then x, y, z are nonzero and pairwise distinct, and $z = x \oplus y$ for this ordering.*

Proof. Equation (4.2) gives $x \oplus y \oplus z = 000$. Therefore

$$x \oplus y = (x \oplus y \oplus z) \oplus z = 000 \oplus z = z.$$

The three values are nonzero because f is nowhere-zero. If $x = y$, then $z = x \oplus y = 000$, a contradiction. If $x = z$, then the flow equation gives $y = 000$, and if $y = z$, it gives $x = 000$. Hence the values are pairwise distinct. $\square$

For completeness, the points of the projective plane over $\mathbb{F}_2$ may be identified with the seven nonzero vectors in $\mathbb{F}_2^3$: each one-dimensional subspace has a unique nonzero vector. Because distinct nonzero vectors x and y are linearly independent, their span is $\{000, x, y, x \oplus y\}$. Its three nonzero vectors are therefore $x, y, x \oplus y = z$, and the corresponding three points form a line in the Fano plane. This explains the name of the lemma; no projective geometry is used later.

5 The triangle at a cubic vertex

Fix a nowhere-zero three-bit flow f . Because G is loopless and cubic, the three members of $\delta(v)$ are three distinct edge objects, even when some of them are parallel in G . At every vertex v , choose an arbitrary ordering a_v, b_v, c_v of these edges. When one vertex is fixed, abbreviate this ordering to a, b, c and put

$$x = f(a), \quad y = f(b), \quad z = f(c) = x \oplus y.$$

By Lemma 4.6, x, y, z are nonzero and pairwise distinct. The identity $z = x \oplus y$ holds for every ordering a, b, c , so the construction does not require a specially chosen order.

For $t \in \{0, 1\}^3$, the map $\tau_t: \{0, 1\}^3 \rightarrow \{0, 1\}^3$ defined by $\tau_t(s) = t \oplus s$ is a bijection, since

$$\tau_t(\tau_t(s)) = t \oplus (t \oplus s) = (t \oplus t) \oplus s = s.$$

Thus τ_t is its own inverse. We call it translation by t . Apply this translation to the three labels $000, x, z$ to obtain

$$t, \quad t \oplus x, \quad t \oplus z. \tag{5.1}$$

The labels $000, x, z$ are distinct by Lemma 4.6, and a bijection preserves distinctness, so the three labels in (5.1) are distinct. Regard them as the vertices of an abstract triangle. For an unordered pair of labels, its *XOR difference* means the XOR of its two elements. XOR difference is invariant under simultaneous translation: for all $p, q, t \in \{0, 1\}^3$,

$$(t \oplus p) \oplus (t \oplus q) = p \oplus q.$$

Assign the three sides to a, b, c as follows:

graph edge	two labels on its triangle side	XOR difference
a	$\{t, t \oplus x\}$	$x = f(a)$
b	$\{t \oplus x, t \oplus z\}$	$x \oplus z = y = f(b)$
c	$\{t, t \oplus z\}$	$z = f(c)$

(5.2)

The first and third rows follow immediately from the definitions of x and z . For the middle row,

$$(t \oplus x) \oplus (t \oplus z) = x \oplus z = x \oplus (x \oplus y) = y. \tag{5.3}$$

Thus the XOR difference of the side assigned to each edge $e \in \delta(v)$ is exactly its flow value $f(e)$. In particular, every side in (5.2) contains two distinct labels because its XOR difference is nonzero.

Lemma 5.1 (Triangle parity). *For every $s \in \{0, 1\}^3$, exactly zero or two of the three side sets in (5.2) contain s .*

Proof. Write

$$\alpha = t, \quad \beta = t \oplus x, \quad \gamma = t \oplus z.$$

The three labels are distinct, and the three side sets in (5.2) are $\{\alpha, \beta\}$, $\{\beta, \gamma\}$, and $\{\gamma, \alpha\}$. Each of α, β, γ therefore belongs to exactly two side sets. Every $s \in \{0, 1\}^3 \setminus \{\alpha, \beta, \gamma\}$ belongs to none of them. These are all eight possible cases because $\{0, 1\}^3$ is the disjoint union of $\{\alpha, \beta, \gamma\}$ and its complement. This proves the assertion. $\square$

For every incidence (v, e) , meaning that $e \in \delta(v)$, define the *local offset* $g_{v,e} \in \{0, 1\}^3$ by

$$g_{v,e} = \begin{cases} 000, & e = a_v, \\ f(a_v), & e = b_v, \\ 000, & e = c_v. \end{cases} \quad (5.4)$$

The three cases are mutually exclusive and exhaustive because a_v, b_v, c_v are distinct and together form $\delta(v)$. These offsets depend only on the fixed flow and the chosen local edge ordering; they do not depend on the translation parameter t .

Lemma 5.2 (Uniform side formula). *For every vertex v , every incident edge $e \in \delta(v)$, and every $t \in \{0, 1\}^3$, the side assigned to e in (5.2) is*

$$Q_{v,e}(t) = \{t \oplus g_{v,e}, t \oplus g_{v,e} \oplus f(e)\}. \quad (5.5)$$

This is a two-element subset of $\{0, 1\}^3$ whose XOR difference is $f(e)$.

Proof. Fix v and again write a, b, c and x, y, z for the local ordering and flow values. If $e = a$, then $g_{v,a} = 000$, and (5.5) is $\{t, t \oplus x\}$. If $e = b$, then $g_{v,b} = x$, and

$$Q_{v,b}(t) = \{t \oplus x, t \oplus x \oplus f(b)\} = \{t \oplus x, t \oplus x \oplus y\} = \{t \oplus x, t \oplus z\}.$$

If $e = c$, then $g_{v,c} = 000$, and (5.5) is $\{t, t \oplus z\}$. These are precisely the three rows of (5.2). Finally, the XOR difference of the two entries in (5.5) is

$$(t \oplus g_{v,e}) \oplus (t \oplus g_{v,e} \oplus f(e)) = f(e),$$

because the two copies of both t and $g_{v,e}$ cancel. This value is nonzero because f is nowhere-zero, so the two entries are distinct. $\square$

The offset belongs to the incidence (v, e) , not to the edge e alone. Consequently, the two endpoints of one edge may assign it different offsets; the next section addresses exactly this compatibility issue.

6 Edge-gluing

Continue with the loopless cubic multigraph G , the nowhere-zero three-bit flow f , and the local edge orderings fixed in Section 5. Each vertex has its own labeled triangle. We seek one translation $t_v \in \{0, 1\}^3$ for every vertex v such that, for every edge $e = uv$, the side assigned to e by the triangle at u is the same unordered two-element set as the side assigned to e by the triangle at v .

For a bit $\epsilon \in \{0, 1\} = \mathbb{F}_2$ and a vector $s \in \{0, 1\}^3$, the notation ϵs means scalar multiplication in the vector space $\mathbb{F}_2^3$. Equivalently, multiplication is performed in each coordinate; thus $0s = 000$ and $1s = s$.

Lemma 6.1 (When two labeled sides agree). *Let $A, B, p \in \{0, 1\}^3$ with $p \neq 000$. Then*

$$\{A, A \oplus p\} = \{B, B \oplus p\}$$

if and only if

$$A \oplus B \in \{000, p\}.$$

When these conditions hold, there is a unique bit $\epsilon \in \{0, 1\}$ such that $A \oplus B = \epsilon p$.

Proof. Because $p \neq 000$, both displayed sets have two elements. If the unordered pairs agree, then A belongs to $\{B, B \oplus p\}$, so $A = B$ or $A = B \oplus p$. Therefore $A \oplus B$ is 000 or p .

Conversely, $A \oplus B = 000$ implies $A = B$, and the pairs agree term by term. If $A \oplus B = p$, then $A = B \oplus p$ and $A \oplus p = B$, so the same two elements occur in the opposite order. Finally, $000 \neq p$, so exactly one of the two possibilities holds. Taking $\epsilon = 0$ in the first case and $\epsilon = 1$ in the second proves existence and uniqueness of ϵ . $\square$

Fix an edge $e = uv$. The standing loopless assumption gives $u \neq v$. Define its *offset discrepancy*

$$d_e = g_{u,e} \oplus g_{v,e}. \quad (6.1)$$

This definition is independent of whether the endpoints are written as u, v or as v, u , because XOR is commutative.

Proposition 6.2 (The edge-gluing equation). *Fix translations $t_v \in \{0, 1\}^3$ at all vertices, and fix an edge $e = uv$. The two endpoint triangles assign the same side set to e if and only if there exists a bit $\epsilon_e \in \{0, 1\}$ such that*

$$t_u \oplus t_v \oplus \epsilon_e f(e) = d_e. \quad (6.2)$$

If such a bit exists, it is unique.

Proof. By Lemma 5.2, the endpoint side sets are

$$\{A_e, A_e \oplus f(e)\} \quad \text{and} \quad \{B_e, B_e \oplus f(e)\},$$

where

$$A_e = t_u \oplus g_{u,e}, \quad B_e = t_v \oplus g_{v,e}.$$

The value $f(e)$ is nonzero. Lemma 6.1, applied with $p = f(e)$, therefore says that the two side sets agree if and only if there is a unique $\epsilon_e \in \{0, 1\}$ for which

$$A_e \oplus B_e = \epsilon_e f(e).$$

Using (6.1), the left side is

$$A_e \oplus B_e = t_u \oplus t_v \oplus g_{u,e} \oplus g_{v,e} = t_u \oplus t_v \oplus d_e.$$

Consequently, side agreement is equivalent to the following chain of equivalent equations:

$$\begin{aligned} t_u \oplus t_v \oplus d_e &= \epsilon_e f(e), \\ t_u \oplus t_v \oplus d_e \oplus \epsilon_e f(e) &= 000, \\ t_u \oplus t_v \oplus \epsilon_e f(e) &= d_e. \end{aligned}$$

The last equation is (6.2). Each passage merely XORs the same vector with both sides and is therefore reversible. The uniqueness of ϵ_e is the uniqueness supplied by Lemma 6.1. $\square$

Equation (6.2) is unchanged if u and v are interchanged, so it requires no orientation of e .

We call the family of equations (6.2), one for every edge, the *triangle-gluing system*. Its unknowns are the three coordinate bits of each t_v and one bit ϵ_e for each edge; the values $f(e)$ and d_e are fixed data. If $f(e) = (f_1(e), f_2(e), f_3(e))$ and similarly for t_v and d_e , then the i th coordinate of (6.2) is

$$(t_u)_i \oplus (t_v)_i \oplus f_i(e)\epsilon_e = (d_e)_i \quad (i = 1, 2, 3).$$

Thus every vector equation is exactly three scalar linear equations over $\mathbb{F}_2$. Applying Proposition 6.2 separately to each edge shows that a family $(t_v)_{v \in V(G)}$ makes all endpoint side sets agree simultaneously if and only if it can be extended by bits $(\epsilon_e)_{e \in E(G)}$ to a solution of the triangle-gluing system. Parallel edges are distinct members of $E(G)$ and therefore contribute distinct equations, each with its own discrepancy d_e and unknown ϵ_e . Hence the vertex triangles can be glued along every edge simultaneously if and only if this binary system has a solution.

The rest of the main proof shows that this system is always consistent.

7 A certificate for inconsistent XOR equations

We now analyze the solvability of the triangle-gluing system from Section 6. We begin with a basic fact about finite binary linear systems.

Lemma 7.1 (XOR inconsistency certificate). *Consider a finite system of linear equations over $\mathbb{F}_2$, written using XOR. Thus every equation has the form*

$$\bigoplus_{j \in J} x_j = b,$$

where the x_j are bit-valued unknowns, J is a set of indices, and $b \in \{0, 1\}$; variables with coefficient zero are omitted, and the XOR over the empty set is 0. The system is inconsistent if and only if the XOR of a subfamily of its equations has every unknown cancel from the left side and has right side equal to 1. Equivalently, an inconsistent system has a certificate of the form $0 = 1$.

Proof. If XORing a subfamily of equations gives $0 = 1$, no assignment can satisfy all of those equations, so the system is inconsistent.

Conversely, write the system as an augmented matrix $[M \mid b]$ over $\mathbb{F}_2$ and perform Gaussian elimination to row-echelon form. Row interchanges merely reorder the equations. Replacing a row R_i by $R_i \oplus R_j$ with $i \neq j$ also preserves the solution set: an assignment satisfies R_i and R_j if and only if it satisfies $R_i \oplus R_j$ and R_j . These are the only nontrivial elementary row operations over $\mathbb{F}_2$.

Initially, every row is the XOR of a one-element subfamily of the original rows. This property is preserved by row interchange, and replacing two such XORs by their XOR corresponds to taking the symmetric difference of their index sets. Hence every row produced by elimination is the XOR of a subfamily of the original rows.

If row-echelon form contains $[0 \cdots 0 \mid 1]$, then it contains the false equation $0 = 1$ and is inconsistent. Conversely, if no such row occurs, set all free variables to zero and solve successively for the pivot variables, starting with the last nonzero row. This gives a solution, so the system is consistent. Therefore an inconsistent system produces a row $[0 \cdots 0 \mid 1]$, and the corresponding XOR of original equations is the required certificate. $\square$

For example, if p and q are bit-valued unknowns, the equations

$$p \oplus q = 0, \quad p \oplus q = 1$$

are inconsistent. XORing them cancels both unknowns and gives $0 = 1$. The certificate lemma says that every inconsistent binary system has a possibly larger certificate of exactly this kind.

Here is the matrix form of the argument that follows. Order the scalar unknowns by collecting the three coordinates of every t_v and then the bits ϵ_e , and write

$$\xi \in \mathbb{F}_2^{3|V(G)|+|E(G)|}$$

for the resulting column vector. Order the three coordinate equations of (6.2) for all edges, concatenate the coordinates of the discrepancies d_e into a column vector

$$\mathbf{d} \in \mathbb{F}_2^{3|E(G)|},$$

and let A be the corresponding coefficient matrix. The triangle-gluing system is then

$$A\xi = \mathbf{d}. \tag{7.1}$$

Lemma 7.1 says equivalently that this system is inconsistent exactly when there is a row-selection vector $y \in \mathbb{F}_2^{3|E(G)|}$ such that

$$A^\top y = 0 \quad \text{and} \quad y^\top \mathbf{d} = 1. \tag{7.2}$$

Consequently, $A\xi = \mathbf{d}$ is solvable if and only if

$$y^\top \mathbf{d} = 0 \quad \text{for every } y \in \ker A^\top. \tag{7.3}$$

The tester notation below is the edge-block form of this criterion: the three coordinates of y belonging to edge e are packaged as one vector $r_e \in \mathbb{F}_2^3$. The rest of the proof gives an elementary description of $\ker A^\top$ and verifies (7.3) without invoking any further duality theorem.

Every vector equation in (6.2) is three scalar bit equations. A convenient way to choose an XOR of some of those three equations is to use a *parity tester*.

Definition 7.2. For a tester $r = (r_1, r_2, r_3) \in \{0, 1\}^3$ and a label $s = (s_1, s_2, s_3) \in \{0, 1\}^3$, define

$$r \cdot s = r_1 s_1 \oplus r_2 s_2 \oplus r_3 s_3 \in \{0, 1\}. \tag{7.4}$$

Here the products $r_i s_i$ are ordinary products of bits, equivalently products in $\mathbb{F}_2$. Applying r to a three-bit equation means taking the XOR of precisely those coordinate equations for which the corresponding bit of r is one. Thus the eight testers in $\{0, 1\}^3$ are in bijection with the eight subsets of the three coordinate equations; the tester 000 selects none of them.

For instance, the tester $r = 101$ selects the first and third coordinate equations, and $101 \cdot 110 = 1 \oplus 0 \oplus 0 = 1$. Directly from (7.4), the dot product is linear in each argument:

$$r \cdot (s \oplus s') = (r \cdot s) \oplus (r \cdot s'), \quad (r \oplus r') \cdot s = (r \cdot s) \oplus (r' \cdot s).$$

Also, if $r \cdot s = 0$ for every $s \in \{0, 1\}^3$, then $r = 000$: taking s to be 100, 010, and 001 reads off the three coordinates of r .

Assign one tester $r_e \in \{0, 1\}^3$ to every edge equation (6.2), apply r_e to that equation, and XOR the resulting scalar equations over all edges. The unknowns cancel precisely under the conditions in the next lemma.

Lemma 7.3 (Shape of an inconsistency certificate). *The triangle-gluing system is inconsistent if and only if there is a family of testers $r_e \in \{0,1\}^3$, one for each edge, such that*

$$r_e \cdot f(e) = 0 \quad (e \in E(G)), \quad (7.5)$$

$$\bigoplus_{e \in \delta(v)} r_e = 000 \quad (v \in V(G)), \quad (7.6)$$

and

$$\bigoplus_{e \in E(G)} r_e \cdot d_e = 1. \quad (7.7)$$

Proof. Expand every equation (6.2) into three scalar equations. By Lemma 7.1, inconsistency gives an XOR of scalar edge equations whose unknown coefficients vanish and whose right side is one. For each edge e , record which of its three coordinate equations were used; their indicator vector is $r_e \in \{0,1\}^3$. The preceding bijection shows that this records the selected scalar equations uniquely. If none of the three equations for e was used, then $r_e = 000$.

Applying r_e to the vector equation (6.2) gives the scalar equation

$$(r_e \cdot t_u) \oplus (r_e \cdot t_v) \oplus \epsilon_e(r_e \cdot f(e)) = r_e \cdot d_e.$$

Indeed, the two linearity identities above handle every XOR term. For the scalar-multiplication term, $r_e \cdot (\epsilon_e f(e)) = \epsilon_e(r_e \cdot f(e))$, as follows by checking the two possible values $\epsilon_e = 0, 1$.

The bit ϵ_e occurs only in the equation for edge e . After tester r_e is applied, its coefficient is $r_e \cdot f(e)$. Cancellation of every ϵ_e is therefore exactly (7.5).

Because G is loopless, the vector t_v occurs once in the equation of every edge incident with v and nowhere else. After all tested equations are XORed, put

$$R_v = \bigoplus_{e \in \delta(v)} r_e \in \{0,1\}^3.$$

The total contribution involving the three coordinate unknowns of t_v is

$$\bigoplus_{e \in \delta(v)} (r_e \cdot t_v) = R_v \cdot t_v = (R_v)_1(t_v)_1 \oplus (R_v)_2(t_v)_2 \oplus (R_v)_3(t_v)_3.$$

Consequently, all three coordinate unknowns $(t_v)_i$ have coefficient zero if and only if $(R_v)_i = 0$ for $i = 1, 2, 3$, equivalently $R_v = 000$. Thus cancellation of all vertex-translation unknowns is exactly (7.6).

After all unknowns cancel, the XOR of the right sides is $\bigoplus_{e \in E(G)} r_e \cdot d_e$. For the resulting equation to be $0 = 1$, this XOR must be one, giving (7.7).

Conversely, suppose testers satisfy (7.5)–(7.7). Apply r_e to the equation for each edge e and XOR the resulting scalar equations. Conditions (7.5) and (7.6) cancel every ϵ_e and every coordinate of every t_v , while (7.7) makes the XOR of the right sides equal to 1. The result is $0 = 1$, so Lemma 7.1 proves that the gluing system is inconsistent. $\square$

Thus compatibility will follow if we prove that testers satisfying (7.5) and (7.6) always give right-side parity zero rather than one.

8 The local tester

Continue with a tester family satisfying the two unknown-cancellation conditions (7.5) and (7.6); condition (7.7) is not assumed. Fix a vertex v . Because G is cubic and loopless, its local ordering a, b, c consists of exactly the three distinct edge objects in $\delta(v)$. As before, put

$$x = f(a), \quad y = f(b), \quad z = f(c) = x \oplus y.$$

Write r_a, r_b, r_c for the testers attached to these three edges. Conditions (7.5) and (7.6) say, respectively,

$$r_a \cdot x = 0, \quad r_b \cdot y = 0, \quad r_c \cdot z = 0, \tag{8.1}$$

and

$$r_a \oplus r_b \oplus r_c = 000. \tag{8.2}$$

Define the bit

$$\lambda_v = r_b \cdot x. \tag{8.3}$$

The local discrepancy contribution at v is

$$\begin{aligned} \bigoplus_{e \in \delta(v)} (r_e \cdot g_{v,e}) &= (r_a \cdot 000) \oplus (r_b \cdot x) \oplus (r_c \cdot 000) \\ &= \lambda_v. \end{aligned} \tag{8.4}$$

Here we used the local offsets $g_{v,a} = 000$, $g_{v,b} = x$, and $g_{v,c} = 000$ from (5.4), together with $r \cdot 000 = 0$ for every tester r .

The next lemma gives a graph-theoretic interpretation of this bit.

Lemma 8.1 (Local tester parity). *Under conditions (8.1) and (8.2),*

$$\lambda_v \equiv \#\{e \in \delta(v) : r_e \neq 000\} \pmod{2}. \tag{8.5}$$

Thus the local discrepancy contribution is the parity of the number of nonzero testers at the vertex.

Proof. The colors x and y are distinct and nonzero by Lemma 4.6. Over $\mathbb{F}_2$, two nonzero vectors are linearly dependent only when they are equal, so x and y are linearly independent. Their span is $\{000, x, y, x \oplus y\}$ and therefore has four elements. Since $\mathbb{F}_2^3$ has eight elements, there is a vector $w \in \{0, 1\}^3$ outside this span. The vector w is not a linear combination of x and y , so x, y, w are linearly independent. Three linearly independent vectors in the three-dimensional space $\mathbb{F}_2^3$ form a basis.

A tester is uniquely determined by its three answers on this basis. Indeed, if two testers have the same answers, their XOR has dot product zero with x, y, w . Linearity then gives dot product zero with every vector in $\mathbb{F}_2^3$, so the nondegeneracy observation following (7.4) says that their XOR is 000. In particular, a tester is 000 if and only if its answer triple on x, y, w is $(0, 0, 0)$.

Define the bits

$$A = r_a \cdot w, \quad B = r_b \cdot w.$$

We determine all answers of the three testers. XORing both sides of (8.2) with r_c gives $r_c = r_a \oplus r_b$. By linearity of the dot product, evaluating this identity at x and using $r_a \cdot x = 0$ gives

$$r_c \cdot x = (r_a \cdot x) \oplus (r_b \cdot x) = \lambda_v.$$

Evaluating at y and using $r_b \cdot y = 0$ gives

$$r_c \cdot y = (r_a \cdot y) \oplus (r_b \cdot y) = r_a \cdot y.$$

Finally, $r_c \cdot z = 0$ and $z = x \oplus y$, so

$$0 = (r_c \cdot x) \oplus (r_c \cdot y) = \lambda_v \oplus (r_a \cdot y).$$

Hence $r_a \cdot y = \lambda_v$. Evaluating $r_c = r_a \oplus r_b$ at w also gives

$$r_c \cdot w = (r_a \cdot w) \oplus (r_b \cdot w) = A \oplus B.$$

On the basis x, y, w , the testers therefore have answer triples

	x	y	w
r_a	0	λ_v	A
r_b	λ_v	0	B
r_c	λ_v	λ_v	$A \oplus B$

(8.6)

If $\lambda_v = 1$, each row has a 1 among its first two entries. Therefore all three testers are nonzero. The number of nonzero testers is three, whose parity is one.

If $\lambda_v = 0$, the first two columns are zero. The testers are nonzero exactly when their last entries $A, B, A \oplus B$ are one, because a tester is zero exactly when all three of its basis answers are zero. The complete list of possible triples of last entries, obtained from the four possible pairs $(A, B) \in \{0, 1\}^2$, is

$$(A, B, A \oplus B) \in \{(0, 0, 0), (0, 1, 1), (1, 0, 1), (1, 1, 0)\}.$$

The first triple has no ones, and each of the other three has two. Hence the number of nonzero testers is even. The cases $\lambda_v = 0$ and $\lambda_v = 1$ exhaust the two possible values of the bit λ_v ; in both cases the parity is λ_v , proving (8.5). $\square$

9 All vertex triangles can be glued

Theorem 9.1 (Triangle compatibility). *Let G be a loopless cubic multigraph with a fixed nowhere-zero three-bit flow f and fixed local edge orderings. Define the offsets $g_{v,e}$ by (5.4) and the discrepancies d_e by (6.1). Then there exist vectors $t_v \in \{0, 1\}^3$, one for each vertex, and bits $\epsilon_e \in \{0, 1\}$, one for each edge, such that*

$$t_u \oplus t_v \oplus \epsilon_e f(e) = d_e \quad \text{for every edge } e = uv.$$

Equivalently, the triangle-gluing system has a solution, and the translated endpoint triangles assign the same side set to every edge.

Proof. Suppose for contradiction that no such vectors and bits exist, so the triangle-gluing system is inconsistent. By Lemma 7.3, choose a tester $r_e \in \{0, 1\}^3$ for every edge e satisfying (7.5)–(7.7). Put

$$S = \{e \in E(G) : r_e \neq 000\}.$$

We evaluate the right-side parity in (7.7).

Let

$$\mathcal{I}(G) = \{(v, e) : v \in V(G), e \in \delta(v)\}$$

be the set of vertex-edge incidences. Because G is loopless, every edge e has two distinct endpoints. For each edge, temporarily denote these endpoints by u_e and v_e in an arbitrary order. Thus e gives exactly the two incidences (u_e, e) and (v_e, e) . By the definition $d_e = g_{u_e, e} \oplus g_{v_e, e}$ and linearity of the dot product,

$$\begin{aligned}
\bigoplus_{e \in E(G)} (r_e \cdot d_e) &= \bigoplus_{e \in E(G)} [(r_e \cdot g_{u_e, e}) \oplus (r_e \cdot g_{v_e, e})] \\
&= \bigoplus_{(v, e) \in \mathcal{I}(G)} (r_e \cdot g_{v, e}) \\
&= \bigoplus_{v \in V(G)} \bigoplus_{e \in \delta(v)} (r_e \cdot g_{v, e}).
\end{aligned} \tag{9.1}$$

Interchanging u_e and v_e only interchanges the two terms inside the brackets. The remaining equalities merely reindex and regroup a finite XOR, so no orientation or ordering of the incidences is involved. Parallel edges remain distinct because the edge object e is part of every incidence.

The chosen tester family satisfies (7.5) and (7.6), so the calculation in Section 8 applies at every vertex, using the fixed local ordering there. Equation (8.4) identifies the inner XOR in (9.1) with λ_v , and Lemma 8.1 gives

$$\begin{aligned}
\bigoplus_{e \in E(G)} (r_e \cdot d_e) &= \bigoplus_{v \in V(G)} \lambda_v \\
&\equiv \sum_{v \in V(G)} \lambda_v \pmod{2} \\
&\equiv \sum_{v \in V(G)} |S \cap \delta(v)| \pmod{2}.
\end{aligned} \tag{9.2}$$

Here XOR is addition modulo two, and the last congruence sums (8.5) over all vertices. Each edge of S is counted twice in (9.2), once at each endpoint; parallel edges remain distinct. Consequently,

$$\sum_{v \in V(G)} |S \cap \delta(v)| = 2|S| \equiv 0 \pmod{2}.$$

The left side of (7.7) is itself a bit, so congruence to zero modulo two forces it to equal 0. This contradicts the assertion in (7.7) that it equals 1. No tester family satisfying (7.5)–(7.7) exists. By the equivalence in Lemma 7.3, the gluing system is consistent and hence has a solution. Proposition 6.2 then gives the equivalent side-set agreement asserted in the theorem. $\square$

The proof uses neither connectedness nor bridgelessness once the nowhere-zero three-bit flow has been fixed; it still uses cubicity in the local calculation and looplessness in counting two distinct endpoint incidences per edge. Lemma 7.3 says that every possible obstruction is a tester certificate, and the incidence count above proves that the right side of every such certificate is zero rather than one.

10 From glued triangles to cycles

Continue with the loopless cubic multigraph G , the fixed nowhere-zero three-bit flow and local edge orderings, and choose a solution (t_v, ϵ_e) supplied by Theorem 9.1. For every incidence (v, e) , set

$$Q_{v, e} = Q_{v, e}(t_v) = \{t_v \oplus g_{v, e}, t_v \oplus g_{v, e} \oplus f(e)\}. \tag{10.1}$$

By Lemma 5.2, this is the triangle side assigned to e at v , and it has two distinct elements. Proposition 6.2 implies

$$Q_{u,e} = Q_{v,e} \quad \text{for every edge } e = uv.$$

Define the common two-element set by

$$P_e = Q_{u,e} = Q_{v,e}. \quad (10.2)$$

Lemma 10.1 (Labels trace cycles). *For a fixed label $s \in \{0, 1\}^3$, let*

$$M_s = \{e \in E(G) : s \in P_e\}.$$

Let $G_s = (V(G), M_s)$ be the spanning subgraph of G with this edge set. Here each member of M_s retains its identity as an edge object, so parallel edges remain distinct. Every vertex has degree zero or two in G_s . Consequently, every connected component of G_s containing an edge is a cycle.

Proof. Fix a vertex v . Its three incident edges are distinct edge objects, even if some are parallel. For every $e \in \delta(v)$, equality (10.2) gives $P_e = Q_{v,e}$, so the three global pairs on the incident edges are exactly the three translated triangle sides at v . By Lemma 5.1, the label s belongs to exactly zero or two of those side sets. By the definition of M_s , this is the number of edge objects of M_s incident with v . Since G_s is loopless, that number is precisely the degree of v in G_s . Thus every vertex has degree zero or two in G_s .

Let K be a connected component of G_s containing an edge. No vertex of K has degree zero in G_s , since such a vertex would be an isolated component rather than a member of the edge-containing component K . Every vertex of K therefore has degree two in G_s . All G_s -edges incident with a vertex of K lie in the same connected component, so its degree in K is also two. Thus K is a connected 2-regular subgraph, which is a cycle by the definition in Section 1. This includes a two-edge cycle formed by a pair of parallel edges. $\square$

Theorem 10.2 (Flow-to-CDC theorem). *Every loopless cubic multigraph that admits a nowhere-zero three-bit flow has a cycle double cover. Moreover, the occurrences in the cover can be indexed by pairs (s, K) , where $s \in \mathbb{F}_2^3$ and K is an edge-containing component of the label subgraph G_s .*

Proof. Fix a nowhere-zero three-bit flow. At every vertex, choose an arbitrary ordering of its three incident edge objects and build the corresponding vertex triangle. Theorem 9.1 supplies a solution to the gluing system, producing the global two-element edge sets P_e defined in (10.2).

For each of the eight labels s , include every edge-containing connected component K of G_s in a multiset $\mathcal{C}$. More precisely, an occurrence of a cycle in $\mathcal{C}$ is indexed by the pair (s, K) . Thus if the same cycle subgraph occurs for two different labels, it contributes two distinct occurrences to the multiset. By Lemma 10.1, every included component is a cycle. The multiset is finite because there are eight labels and each finite graph G_s has finitely many connected components.

Fix an edge e . Its two-element set P_e contains two distinct labels, say s and s' . From the definition of M_q , for $q \in \{0, 1\}^3$, the edge e belongs to M_q if and only if $q \in P_e$. Hence e belongs to exactly the two spanning subgraphs G_s and $G_{s'}$. In each it belongs to exactly one connected component. Those components contain e , so both are included in $\mathcal{C}$, giving one occurrence indexed by s and one indexed by s' . These remain two occurrences even if the two component subgraphs are equal. For every other label $q \notin \{s, s'\}$, the edge e is not in M_q , so no other member of $\mathcal{C}$ contains e . Therefore every edge occurs in exactly two members of $\mathcal{C}$, counted with multiplicity, and $\mathcal{C}$ is a cycle double cover. $\square$

Corollary 10.3 (Cubic conclusion of the argument). *Every loopless bridgeless cubic multigraph has a cycle double cover.*

Proof. Corollary 4.5 supplies a nowhere-zero three-bit flow, so Theorem 10.2 applies. $\square$

The indexing in Theorem 10.2 uses eight possible labels, not necessarily eight cycles. For one label s , the subgraph G_s may have several edge-containing components, and each such component contributes a separate cycle occurrence indexed by (s, K) .

11 Concluding remarks

Our proof has four stages. First, Proposition 3.1 reduces the problem for finite bridgeless multigraphs to the loopless bridgeless cubic case. Its construction works componentwise, removes and later restores loops, expands every nonisolated vertex into a replacement cycle, and projects a cycle double cover of the resulting cubic multigraph back to the original graph.

Second, Corollary 4.5, obtained from the theorems of Seymour and Tutte, supplies a nowhere-zero $\mathbb{F}_2^3$ -flow on the cubic graph. Third, Sections 5–9 convert that flow into locally labeled vertex triangles and prove that the triangles can be translated so that their side-label sets agree across every edge. This is the content of Theorem 9.1. Fourth, Section 10 extracts cycles from the compatible labels and proves the flow-to-cover result, Theorem 10.2. Applying the three-bit flow input gives the cubic conclusion, Corollary 10.3.

The main compatibility step is a finite binary-equation argument. Proposition 6.2 converts side agreement into the triangle-gluing system. Lemma 7.3 shows that an inconsistent system would have edge testers satisfying (7.5)–(7.7). Lemma 8.1 identifies each vertex contribution with the parity of its incident nonzero testers. Theorem 9.1 then counts every nonzero edge tester once at each endpoint, forcing the certificate’s right-side parity to be zero rather than one.

Once compatibility is established, each of the eight three-bit labels selects a subgraph of degree zero or two at every vertex. Its edge-containing components are cycles, and the two distinct labels on each edge place that edge in exactly two indexed cycle components.

Theorem 11.1 (Cycle Double Cover Theorem). *Every finite bridgeless multigraph has a cycle double cover.*

Proof. Corollary 10.3 establishes the hypothesis of Proposition 3.1: every loopless bridgeless cubic multigraph has a cycle double cover. Applying that proposition gives a cycle double cover of every finite bridgeless multigraph.

For clarity, all exceptional cases are already included in the proposition. Its proof works separately in disconnected components, ignores isolated vertices, and handles a graph with no non-loop edges directly. In the remaining case it constructs a loopless bridgeless cubic multigraph, applies Corollary 10.3, and discards cover cycles that use no old edge. It projects every remaining cover cycle to a closed trail, decomposes that trail into cycles, and finally restores every original loop by inserting two copies of that one-edge cycle. Thus no connectedness, looplessness, or minimum-degree assumption is being added to the theorem stated here. $\square$

Acknowledgements: A high level four-stage strategy was written by the author and provided to Codex and asked it to apply the strategy to open problems related to the Cycle Double Cover Conjecture (CDCC). Codex listed eight related open problems. The author used multiple rounds of prompting with some graph theoretic ideas and nudging Codex in the right direction by making it work out the steps on several examples of graphs, fill the proof gaps and fix the errors. The

author then asked Codex to analyze the previously generated proofs and find missing steps. In later rounds of prompting, the author provided more graph theoretic details and iteratively asked Codex to verify all the steps section by section.

During the same week, OpenAI also announced a proof of CDCC [12]. There are differences between our proof and OpenAI’s proof. OpenAI’s proof uses an abstract map and its annihilator in the dual space. We use scalar XOR equations, Gaussian elimination certificates, dot products and tester vectors. OpenAI’s proof uses nowhere-zero 8-flow results of Kilpatrick and Jaeger and Tutte’s group-flow equivalence. Our proof uses Seymour 6-flow as a starting point. Both these flows finally produce the same objects. Both papers share the high-level *flow to label* arguments. OpenAI’s proof is linear-algebraic while our proof is graph-theoretic and combinatorial in nature.

We believe these techniques (triangles, edge gluing, inconsistency certificates, cycle extraction) might be useful to prove other related conjectures including Strong Cycle Double Cover Conjecture.

References

- [1] B. Alspach, L. A. Goddyn, and C.-Q. Zhang, *Graphs with the circuit cover property*, Trans. Amer. Math. Soc. 344 (1994), 131–154.
- [2] B. Ghanbari and R. Šámal, *Facial diagrams and cycle double cover*, arXiv:2605.01410 (2026), arXiv:2605.01410.
- [3] A. Itai and M. Rodeh, *Covering a graph by circuits*, in Automata, Languages and Programming, Lecture Notes in Comput. Sci. 62, Springer, 1978, 289–299.
- [4] F. Jaeger, *A survey of the cycle double cover conjecture*, in Cycles in Graphs, North-Holland Math. Stud. 115, North-Holland, 1985, 1–12.
- [5] P. D. Seymour, *Sums of circuits*, in Graph Theory and Related Topics, Academic Press, 1979, 341–355.
- [6] P. D. Seymour, *Nowhere-zero 6-flows*, J. Combin. Theory Ser. B 30 (1981), 130–135, doi:10.1016/0095-8956(81)90058-7.
- [7] G. Szekeres, *Polyhedral decompositions of cubic graphs*, Bull. Austral. Math. Soc. 8 (1973), 367–387.
- [8] W. T. Tutte, *A contribution to the theory of chromatic polynomials*, Canad. J. Math. 6 (1954), 80–91, doi:10.4153/CJM-1954-010-9.
- [9] D. Xie and C.-Q. Zhang, *Flows, flow-pair covers and cycle double covers*, Discrete Math. 309 (2009), 4682–4689, doi:10.1016/j.disc.2008.05.056.
- [10] C.-Q. Zhang, *Nowhere-zero 4-flows and cycle double covers*, Discrete Math. 154 (1996), 245–253, doi:10.1016/0012-365X(95)00047-Z.
- [11] C.-Q. Zhang, *Circuit Double Cover of Graphs*, Cambridge University Press, 2012.
- [12] OpenAI, *A PROOF OF THE CYCLE DOUBLE COVER CONJECTURE* https://cdn.openai.com/pdf/04d1d1e4-bc75-476a-97cf-49055cd98d31/cdc_proof.pdf